Abhishek Jha

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Machine learning engineer in the final year of IIT Madras's four-year BS in Data Science and Applications (B.Sc. completed 2026). I build ML systems and LLM agents and take them from training to production: deep learning for vision and medical imaging, multi-agent workflows in LangGraph, RAG, MCP tools, LLM evaluation and voice AI, shipped with FastAPI, Next.js, Docker and CI/CD. Previously Founding Engineer at Trackleo Technologies, where I helped launch the company's first AI platform in under three months. I built Pips AI solo in 15 days.

Education

Indian Institute of Technology (IIT) Madras — **Chennai, India**
BS in Data Science and Applications (4-year, final year) — *Sep 2023 - Expected 2027*
B.Sc. completed in May 2026 after three years; now in the fourth (BS) year. Machine learning, deep learning, statistics, programming, software engineering and MLOps. Includes Diplomas in Programming and in Data Science.

Awards & Certifications

- **Kaggle**: Rank 1 in Apple Disease Classification (IIT Madras course competition); top 1% in Engage 2: Value from Clicks to Conversions.
- **S grade** (highest, 10/10) at IIT Madras in Machine Learning Practice and Modern Application Development I and II; **100/100** on the MLOps project.
- Diploma in Programming, Diploma in Data Science, Advanced Certificate in ML & Data Science (IIT Madras); Gen AI to Agentic AI, NLP, Deep Learning, ML, Python, SQL, Math & Statistics (Codebasics); all 19 online.
- **Languages**: English (fluent), Hindi (native).

Experience

Trackleo Technologies Pvt. Ltd. — **Chennai, India**
Founding Engineer (Web Developer & Data Scientist) — *Feb 2026 - Jul 2026*
- Engineered a deep learning fundus image classifier for five retinal disease classes: medical image preprocessing, feature engineering and model optimization.
- Partnered with the CEO to launch the company's first AI-powered web platform in under three months.
- Owned model experimentation, deployment workflows and platform integration across two core AI initiatives (Python, SQL, FastAPI, Docker, MLflow, CI/CD, Kubernetes).

AtliQ Technologies — **Remote**
Virtual Data Scientist Intern — *Dec 2025 - Jan 2026*
- Built a CNN and ResNet50 fruit freshness classifier (8 fruit types, 97.8% validation accuracy) served through Streamlit, cutting manual inspection from hours to minutes.

Sudha Gopalakrishnan Brain Centre, IIT Madras — **Chennai, India**
AI LLM Intern — *Nov 2025*
- Built 11 MCP tools in Python that let an LLM drive 3D Slicer in natural language for brain MRI visualization and analysis; an estimated 60% time saving.

Web Karo India — **Remote**
AI Product Engineering Intern — *Jan 2025 - Sep 2025*
- Architected WorkSphere AI, a collaborative productivity platform (Kanban, calendar, notes, whiteboard, AI assistants, voice commands) in Next.js, TypeScript, Prisma and PostgreSQL; estimated 35% task-efficiency gain.
- Delivered ML pipelines, backend services, API integrations and automation features.

Skills

- **Machine learning**: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, OpenCV, YOLO, Whisper; feature engineering, model evaluation, hyperparameter tuning.
- **LLMs & agents**: LangGraph, LangChain, RAG (FAISS, ChromaDB, Qdrant), MCP, OpenAI, Claude and Gemini APIs, DeepEval.
- **MLOps & cloud**: MLflow, DVC, Feast, Docker, Kubernetes, GitHub Actions, GCP Vertex AI, AWS, Databricks, Prometheus + Grafana.
- **Backend & web**: FastAPI, Flask, PostgreSQL, Redis, Celery, WebSockets, Next.js, React, React Native/Expo, Supabase.
- **Languages & tools**: Python, SQL, TypeScript, JavaScript; Git, Claude Code, Codex, Cursor, GitHub Copilot, Jupyter, Power BI.

Projects

Products

- **Pips AI**: trading platform with a live rule-compliance view and an AI coach that reviews trades against prop-firm rules. Built solo in 15 days. *FastAPI, PostgreSQL, Next.js, Claude API, Docker* ▪ pipsai.online
- **ShelfScan (so-app)**: startup project pre-incubated and funded under IIT Madras Nirmaan (Jul 2026 - present): scan-and-go supermarket checkout with QR scanning and Razorpay payments. *Expo/React Native, TypeScript, Supabase (RLS, OTP, edge functions), Jest, CI*

AI agents & LLM systems

- **Data Firefighter**: 11-node LangGraph agent that traces a data incident through DataHub lineage to its root cause, proposes a fix and opens a GitHub PR after human approval (DataHub agent hackathon). *LangGraph, FastAPI, Next.js* ▪ GitHub
- **Butterfly Director AI**: 11 specialised agents turn one story premise into branching generated short films. *LangGraph, FastAPI, PostgreSQL, Celery, Redis, Qwen, Wan, CosyVoice* ▪ GitHub
- **LLM Evaluation Platform**: benchmarks LLMs across providers on faithfulness, relevancy, bias and toxicity, with prompt regression tests and a blind Elo arena. *LiteLLM, DeepEval, FastAPI, Celery, Prometheus, Grafana* ▪ GitHub
- **Voice Agents Study**: IIT Madras DS & AI Lab team of 4 (ongoing): enquiry and lead-qualification voice agent with CRM updates, plus a single- vs multi-agent comparison on correctness, latency and cost. *LiveKit, Claude API, FastAPI, PostgreSQL*
- **EchoFlow**: low-latency voice-to-voice assistant with voice activity detection and a swappable STT, LLM and TTS pipeline. *FastAPI, WebSockets, Next.js, Gemini, OpenAI* ▪ GitHub
- **Facility Trust Desk**: evidence-backed AI assessments of what 1,000+ healthcare facilities claim, with human review (Databricks Apps & Agents for Good Hackathon 2026). *OpenAI, Databricks, Streamlit* ▪ GitHub
- **Autonomous Incident Commander**: turns Splunk alerts into an incident timeline, root cause with a confidence score and a PDF report (Splunk hackathon). *MCP, FastAPI, Claude API, React* ▪ GitHub
- **Career Agent AI**: ingests jobs from ATS feeds and crawlers, ranks fit, parses resumes and generates tailored resumes and cover letters. *FastAPI, PostgreSQL, Playwright, Celery, Next.js* ▪ GitHub
- **AgentTrust**: trust layer for AI agents: identities, permissions, an LLM audit panel and an audit trail on Solana devnet. *TypeScript, Next.js, PostgreSQL, Solana* ▪ GitHub
- **LLM apps**: RAG FAQ chatbot, career-guidance chatbot, research-paper explainer and structured-output review analysers. *LangChain, OpenAI, FAISS, Pydantic, Streamlit* ▪ GitHub

Machine learning, vision & speech

- **Long-Video Activity Detection**: detects and localises 10 activity types in long, untrimmed CCTV video, with measured results and failure analysis. *PyTorch, YOLO11n, ByteTrack, OpenCV, Streamlit* ▪ GitHub
- **MLOps Pipelines**: DVC, Feast and MLflow pipelines with CI/CD, deployed on Kubernetes and Vertex AI; 100/100 at IIT Madras. *DVC, Feast, MLflow, Docker, Kubernetes, GCP* ▪ GitHub
- **IndicSpeechLab**: Indian-language ASR evaluation with WER, CER and real-time factor. *faster-whisper, FFmpeg, Streamlit* ▪ GitHub
- **Baaton Se Gaana**: turns a conversation in any of 9 Indian languages into a personalised song. *Sarvam AI, OpenAI, Streamlit* ▪ GitHub
- **System-Threat Prediction**: IIT Madras MLP project (S grade): feature engineering, class imbalance handling and tuned gradient boosting, evaluated on AUC-ROC and F1. *scikit-learn, XGBoost, LightGBM, Optuna* ▪ GitHub
- **Beverage Price Range Predictor**: survey-data model comparison tracked in MLflow, served as a live app. *scikit-learn, XGBoost, MLflow, Streamlit* ▪ GitHub
- **Customer Churn Prediction**: ANN vs XGBoost on telecom and banking data, compared on precision, recall and F1. *TensorFlow/Keras, XGBoost* ▪ GitHub

Full-stack, web3 & data

- **Trekking Management System**: role-based web app for trek scheduling and bookings, with caching and background jobs. *Vue.js, Flask, JWT, Redis, Celery* ▪ GitHub
- **Parking Management SPA**: IIT Madras MAD-1 and MAD-2 projects (S grade): reservations, dashboards, async exports and email reports. *Flask, Vue.js, Redis, Celery* ▪ GitHub
- **RemitAI**: compares stablecoin remittance routes across Ethereum and Layer-2 networks against traditional providers (hackathon MVP). *Next.js, FastAPI, Solidity, Hardhat* ▪ GitHub
- **Notes Apps**: full-stack CRUD apps in MERN and in Next.js with MongoDB. *MongoDB, Express, React, Node.js, Next.js* ▪ GitHub
- **BDM Capstone**: analysed five months of real data for an iron-trading MSME: labour at 30-43% of revenue, 40%+ of customers paying late. *Excel, statistics* ▪ GitHub

Other prototypes

AI Scientist, STEMLab AI, Parallel AI, Future Self AI, ShadowGov AI, TradePulse AI, AI Construction Commander, Lumora and a WhatsApp AI assistant. All are on GitHub.